

Funk volume and Kalai's flag conjecture

Proof

Let $n \geq 1$. All volumes and integrals use ordinary Lebesgue measure. For a convex body $K \subset \mathbb{R}^n$ with $K = -K$ and $0 \in \text{int } K$, define

$$C^\circ = \{y \in \mathbb{R}^n : \langle y, z \rangle \leq 1 \text{ for every } z \in C\}, \quad V_K(\tau) = \int_{\tau K} \text{vol}_n((K - x)^\circ) dx.$$

No Euclidean unit-ball normalization is included in V_K .

Conjecture 1. Symmetric Funk volume

For every symmetric convex body $K \subset \mathbb{R}^n$ and $0 < \tau < 1$,

$$V_K(\tau) \geq \frac{(4 \operatorname{artanh} \tau)^n}{n!}.$$

A complete flag of an n -dimensional polytope P is a chain $F_0 \subsetneq \cdots \subsetneq F_{n-1}$ of nonempty proper faces with $\dim F_i = i$. Appending P to every chain does not change their number.

Conjecture 2. Kalai's flag conjecture

Every centrally symmetric n -dimensional convex polytope P satisfies

$$\# \text{Flags}(P) \geq 2^n n!.$$

Theorem 1. Funk volume and complete flags

Conjectures 1 and 2 hold. In particular, for every symmetric convex body $K \subset \mathbb{R}^n$ and every $0 < \tau < 1$,

$$\frac{(4 \operatorname{artanh} \tau)^n}{n!} \leq V_K(\tau) < \infty,$$

and every symmetric n -dimensional polytope has at least $2^n n!$ complete flags. The volume constant is attained by the cube.

The proof first constructs a holomorphic lens whose boundary density matches translated-polar simplex volumes. A holomorphic mass estimate then supplies the boundary probability inequality. Almost-everywhere packing and convex approximation prove the volume bound. Finally, a flag upper estimate yields the complete-flag inequality.

1. Finiteness and normalization

Lemma 2. Finiteness and linear invariance

For $0 < \tau < 1$, the quantity $V_K(\tau)$ is finite. If $A \in \text{GL}_n(\mathbb{R})$, then $V_{AK}(\tau) = V_K(\tau)$.

Proof. Choose $r > 0$ with $rB_2^n \subset K$. If $x = \tau X$ and $X \in K$, convexity and symmetry give

$$(1 - \tau)K \subset K - \tau X, \quad (K - \tau X)^\circ \subset \frac{1}{r(1 - \tau)}B_2^n.$$

Thus the integrand has a uniform finite bound on the bounded domain τK . It is measurable: the relation $\langle y, z - x \rangle \leq 1$ for all $z \in K$ is closed in (x, y) , so integration of its indicator in y gives a measurable function of x . This also justifies the nonnegative integrals below.

For invariance, $(AK - Ax)^\circ = A^{-T}(K - x)^\circ$. The polar-volume factor $|\det A|^{-1}$ cancels the change-of-variables factor $|\det A|$. $\square$

Lemma 3. The cube attains the stated constant

For $Q = [-1, 1]^n$ and $0 < \tau < 1$,

$$V_Q(\tau) = \frac{(4 \operatorname{artanh} \tau)^n}{n!}.$$

Proof. For $x \in (-1, 1)^n$, the polar of $Q - x$ is the convex hull of

$$\frac{e_i}{1 - x_i}, \quad \frac{-e_i}{1 + x_i} \quad (1 \leq i \leq n).$$

Its intersection with each coordinate orthant is a simplex with vertex 0. These 2^n simplices have disjoint interiors. Consequently,

$$\operatorname{vol}_n((Q - x)^\circ) = \frac{1}{n!} \prod_{i=1}^n \left(\frac{1}{1 - x_i} + \frac{1}{1 + x_i} \right).$$

Tonelli's theorem separates the integral over $[-\tau, \tau]^n$, and

$$\int_{-\tau}^{\tau} \left(\frac{1}{1 - t} + \frac{1}{1 + t} \right) dt = 2 \log \frac{1 + \tau}{1 - \tau} = 4 \operatorname{artanh} \tau.$$

Multiplying these n equal factors proves the claim. $\square$

The radius parameter used later is $\tau_R = 1 - e^{-R}$, for which

$$4 \operatorname{artanh} \tau_R = 2 \log(2e^R - 1), \quad \lim_{R \rightarrow \infty} \frac{4 \operatorname{artanh} \tau_R}{R} = 2. \quad (1)$$

2. The explicit holomorphic lens

Fix $0 < \tau < 1$ and set

$$b = \frac{2 \operatorname{artanh} \tau}{\pi}, \quad c = \sqrt{1 - \tau^2}, \quad \kappa = \frac{4b}{\pi\tau}, \quad F(z) = \kappa \sum_{j=0}^{\infty} \frac{(-1)^j z^{2j+1}}{(2j+1)(2j+1-ib)}.$$

Lemma 4. Lens geometry

The map F is holomorphic on $\mathbb{D}$, continuous on $\overline{\mathbb{D}}$, and maps $\mathbb{D}$ biholomorphically onto the interior of a symmetric convex body $L_\tau \subset \mathbb{C}$. It maps $\partial\mathbb{D}$ homeomorphically onto ∂L_τ . On the right semicircle its height is

$$T(\theta) = \Im F(e^{i\theta}) = \frac{1 - ce^{-b\theta}}{\tau}, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}. \quad (2)$$

The two boundary branches, written as $f_\pm(t) + it$, have nonzero second derivatives for $-1 < t < 1$.

Proof. The coefficients are bounded in modulus by $(2j+1)^{-2}$, up to the constant κ . Uniform convergence on $\overline{\mathbb{D}}$ gives continuity, and termwise differentiation inside the disk yields

$$zF'(z) - ibF(z) = \kappa \arctan z.$$

Away from $z = \pm i$ this differential equation extends the boundary derivatives. On the open right semicircle, $\Re \arctan(e^{i\theta}) = \pi/4$ and $\frac{d}{d\theta} \arctan(e^{i\theta}) = i/(2 \cos \theta)$. Writing $F(e^{i\theta}) = x(\theta) + iT(\theta)$ gives

$$T' + bT = b/\tau, \quad x'' + bx' = -\frac{\kappa}{2 \cos \theta}.$$

Oddness of F implies $T(-\pi/2) = -T(\pi/2)$. Solving the first equation with this relation gives (2), since $c = 1/\cosh(b\pi/2)$. In particular $T(\pm\pi/2) = \pm 1$ and $T' > 0$.

Put $\alpha = \Re F(i)$ and $Q(\theta) = x(\theta) - \alpha T(\theta)$. Then $Q(\pm\pi/2) = 0$ and $(e^{b\theta}Q')' = -\kappa e^{b\theta}/(2 \cos \theta) < 0$. A negative interior minimum would have $Q' = 0$ and $Q'' \geq 0$, contradicting this equation; the same argument excludes an interior zero. Thus $Q > 0$ inside. Define $q(T(\theta)) = Q(\theta)$. Then

$$q''(T(\theta)) = -\frac{\kappa}{2 \cos \theta T'(\theta)^2} < 0.$$

The opposite branch is obtained by oddness, so the boundary is the simple closed curve of the convex body

$$L_\tau = \{x + it : -1 \leq t \leq 1, -q(-t) \leq x - \alpha t \leq q(t)\}.$$

Here $q(\pm 1) = 0$ and $q > 0$ inside; hence $0 \in \operatorname{int} L_\tau$. For any point off the boundary, uniform convergence of $F(re^{i\theta})$ as $r \uparrow 1$ identifies its winding number with that of this Jordan curve. The argument principle counts zeros with multiplicity. The count at 0 is positive, so the interior winding number is 1; outside it is 0. Thus every interior point has exactly one simple preimage, proving biholomorphy. Finally $f_+(t) = \alpha t + q(t)$ and $f_-(t) = \alpha t - q(-t)$ have strictly negative and positive second derivatives, respectively. $\square$

3. Boundary probability and polar simplices

Let μ_τ be the image of $d\theta/(2\pi)$ under $\theta \mapsto F(e^{i\theta})$. Let $a_1, \dots, a_m \in \mathbb{R}^n$ give a bounded strip body

$$P = \{X : |\langle a_j, X \rangle| \leq 1 \ (1 \leq j \leq m)\}.$$

Assume the presentation is *signed irredundant*: no a_j lies in $\text{conv}\{\pm a_\ell : \ell \neq j\}$. Every symmetric full-dimensional polytope has such a presentation, by taking one vertex from each opposite pair of vertices of its polar.

For an increasing n -element selection $I = (i_1, \dots, i_n)$ with invertible row matrix B_I , and $\varepsilon \in \{\pm 1\}^n$, set

$$Y_{I,\varepsilon}(X) = B_I^{-1}(f_{\varepsilon_k}(\langle a_{i_k}, X \rangle))_{k=1}^n.$$

Let $S_{I,\varepsilon} \subset P$ consist of those X for which $\langle a_j, Y_{I,\varepsilon}(X) \rangle + i\langle a_j, X \rangle \in L_\tau$ for all j . These are measurable sets, by continuity of the displayed maps.

Lemma 5. Exact probability-volume identity

The joint density of branch ε and height t under μ_τ is

$$\rho_\varepsilon(t) = \frac{\tau}{4 \operatorname{artanh} \tau (1 - \varepsilon \tau t)}, \quad -1 < t < 1.$$

For $X \in P$, define

$$\Delta_{I,\varepsilon}(X) = \text{conv} \left(0, \frac{\varepsilon_k a_{i_k}}{1 - \varepsilon_k \tau \langle a_{i_k}, X \rangle} \ (1 \leq k \leq n) \right).$$

Writing $E_I = \{w \in (\partial L_\tau)^n : \langle a_j, B_I^{-1} w \rangle \in L_\tau \ \forall j\}$, one has

$$\mu_\tau^{\otimes n}(E_I) = n! \left(\frac{\tau}{4 \operatorname{artanh} \tau} \right)^n \sum_{\varepsilon} \int_{S_{I,\varepsilon}} \text{vol}_n(\Delta_{I,\varepsilon}(X)) dX. \quad (3)$$

Proof. On the right branch, $T' = (b/\tau)(1 - \tau T)$, so $d\theta/(2\pi) = \rho_+(t)dt$. Oddness gives $\rho_-(t) = \rho_+(-t)$. Each branch has mass 1/2, and the endpoints have mass zero.

For fixed signs, use $t = B_I X$. Its Jacobian is $|\det B_I|$. The interpolant is exactly $Y_{I,\varepsilon}(X) + iX$, and its feasibility is exactly $X \in S_{I,\varepsilon}$. The simplex determinant gives

$$\text{vol}_n(\Delta_{I,\varepsilon}(X)) = \frac{|\det B_I|}{n! \prod_k (1 - \varepsilon_k \tau \langle a_{i_k}, X \rangle)}.$$

All denominators are positive. Substitution of the densities proves (3), after summing the disjoint branch events. No independence between branch and height is used. $\square$

4. Almost-everywhere packing

Lemma 6. Packing inequality

For the signed irredundant presentation above,

$$\sum_I \mu_\tau^{\otimes n}(E_I) \leq \frac{n!}{(4 \operatorname{artanh} \tau)^n} V_P(\tau). \quad (4)$$

Proof. Each vertex $v = \varepsilon a_j / (1 - \varepsilon \tau \langle a_j, X \rangle)$ belongs to $(P - \tau X)^\circ$: for $p \in P$,

$$\langle v, p - \tau X \rangle = \frac{\varepsilon \langle a_j, p \rangle - \varepsilon \tau \langle a_j, X \rangle}{1 - \varepsilon \tau \langle a_j, X \rangle} \leq 1.$$

Thus all the displayed simplices lie in that polar.

First compare two feasible choices with distinct real witnesses Y, Z . The branch point selected at Y is the appropriate horizontal endpoint of its slice of L_τ . Hence its corresponding polar vertex satisfies $\langle v, Y - Z \rangle \geq 0$. Vertices selected at Z satisfy the opposite inequality. The nonzero functional $\langle \cdot, Y - Z \rangle$ separates the two full-dimensional simplices. Their intersection has zero volume.

It remains to exclude coincident witnesses for distinct choices, outside a null set of X . Distinct signs on the same basis give distinct witnesses whenever all heights are strict. Distinct bases at one witness force an extra active row a_j beyond the first basis. Write $a_j = \sum_k d_k a_{i_k}$. Signed irredundancy implies at least two d_k are nonzero: a one-coordinate representation would make one of the two parallel rows belong to the signed hull of the other rows. Also no row is zero.

In basis height coordinates $t = B_I X$, activity on branch δ requires

$$H(t) = \sum_k d_k f_{\varepsilon_k}(t_k) - f_\delta(d \cdot t) = 0.$$

On the open domain of strict heights, choose $r \neq s$ with $d_r d_s \neq 0$. Lemma 4 gives

$$\partial_s \partial_r H(t) = -d_r d_s f''_\delta(d \cdot t) \neq 0.$$

The portion of $\{H = 0\}$ where $\partial_r H \neq 0$ is locally a C^1 hypersurface. The remaining portion lies in $\{\partial_r H = 0\}$, which is also locally a C^1 hypersurface, because its s -derivative is nonzero. A countable collection of such charts shows that both portions are null. The invertible map B_I preserves nullity. Endpoint heights lie in finitely many hyperplanes. There are finitely many row, basis and sign choices, so their exceptional sets have null union.

For almost every $X \in P$, the feasible simplices consequently have pairwise disjoint interiors. Their boundaries are null, and therefore

$$\sum_{I, \varepsilon} \mathbf{1}_{S_{I, \varepsilon}}(X) \operatorname{vol}_n(\Delta_{I, \varepsilon}(X)) \leq \operatorname{vol}_n((P - \tau X)^\circ).$$

Integrate, use (3), and substitute $x = \tau X$. The factor τ^n cancels, proving (4). $\square$

5. Holomorphic mass at an isolated zero

For a real smooth function on $\mathbb{C}^n$, use

$$d^c = \frac{i}{2}(\bar{\partial} - \partial), \quad dd^c = i\partial\bar{\partial}, \quad (dd^c v)^n = 2^n n! \det(v_{\bar{z}_i z_j}) dV_{2n}.$$

The orientation is $dx_1 \wedge dy_1 \wedge \cdots \wedge dx_n \wedge dy_n$.

Lemma 7. Holomorphic mass estimate

Let $U \subset \mathbb{C}^n$ be open with $0 \in U$, and let $f : U \rightarrow \mathbb{C}^m$ be holomorphic. Assume $f^{-1}(0) = \{0\}$ and

$$f(z) = H(z) + O(|z|^{k+1}), \quad k \geq 1,$$

where H is homogeneous of degree k and has no nonzero common zero. Assume also that $\{z \in U : |f(z)|^2 \leq s\}$ is compact in U for every $0 < s < 1$. Then

$$\int_{\{|f|^2 < 1\}} \sum_{|I|=n} |\det Df_I(z)|^2 dV_{2n}(z) \geq \frac{\pi^n k^n}{n!}. \quad (5)$$

Proof. Write $u = |f|^2$. The Hermitian Hessian of u is $(Df)^* Df$. Away from 0, that of $\log u$ is positive semidefinite because its value on a complex direction ξ is

$$\frac{|f|^2 |Df \xi|^2 - |\langle Df \xi, f \rangle|^2}{|f|^4} \geq 0.$$

Fix a regular value $s \in (0, 1)$ and put $U_s = \{u < s\}$. Its closure is a compact smooth manifold with boundary. Set $\eta = d^c \log u$. On ∂U_s , the pullback of du is zero, and therefore

$$s^{-n} \int_{U_s} (dd^c u)^n = \int_{\partial U_s} \eta \wedge (d\eta)^{n-1}.$$

This follows from Stokes and the identity $dd^c \log u = u^{-1} dd^c u - u^{-2} du \wedge d^c u$. For a small closed ball $\bar{B}_r \subset U_s$, another application of Stokes gives

$$s^{-n} \int_{U_s} (dd^c u)^n \geq \int_{\partial B_r} \eta \wedge (d\eta)^{n-1}, \quad (6)$$

since the difference is the integral of the nonnegative form $(dd^c \log u)^n$ over $U_s \setminus \bar{B}_r$.

On an annulus around the unit sphere, the holomorphic Taylor expansion implies $r^{-2k} u(rz) \rightarrow T(z) := |H(z)|^2$ in C^2 . The function T is strictly positive there. Pullback under $z \mapsto rz$ and continuity of the logarithmic derivatives thus show that the right side of (6) tends to

$$\int_{S^{2n-1}} \alpha \wedge (d\alpha)^{n-1}, \quad \alpha = d^c \log T|_{S^{2n-1}}.$$

We calculate this integral next.

6. The homogeneous flux and its normalization

Proof of Lemma 7, continued. On $S = S^{2n-1}$ define

$$\alpha_0 = d^c \log |z|^2|_S, \quad \beta = d^c (\log T - k \log |z|^2)|_S, \quad \alpha_t = k\alpha_0 + t\beta.$$

Let $W(z) = iz$ be the tangent field generating common phase rotation. Homogeneity implies that $\log T - k \log |z|^2$ is invariant both under positive dilation and under this rotation. Since $d^c v(\xi) = -\frac{1}{2} dv(i\xi)$, it follows that

$$\beta(W) = 0, \quad \alpha_0(W) = 1, \quad \mathcal{L}_W \alpha_t = 0.$$

Cartan's formula now gives $\iota_W d\alpha_t = \mathcal{L}_W \alpha_t - d(\alpha_t(W)) = 0$. Consequently the top-degree form $\beta \wedge (d\alpha_t)^{n-1}$ on S has zero contraction with a nowhere-zero tangent field, and hence vanishes identically.

Differentiation under the integral and integration by parts on the closed sphere give

$$\frac{d}{dt} \int_S \alpha_t \wedge (d\alpha_t)^{n-1} = n \int_S \beta \wedge (d\alpha_t)^{n-1} = 0.$$

For $n = 1$ the same equality is just differentiation of the integral of a one-form. Evaluating at $t = 0$ and $t = 1$ yields

$$\begin{aligned} \int_S \alpha \wedge (d\alpha)^{n-1} &= k^n \int_S \alpha_0 \wedge (d\alpha_0)^{n-1} \\ &= k^n \int_{B_1} (dd^c |z|^2)^n \\ &= k^n 2^n n! \operatorname{vol}_{2n}(B_1) = (2\pi k)^n. \end{aligned}$$

In the middle line, the restrictions of $d^c \log |z|^2$ and $dd^c \log |z|^2$ to S equal those of $d^c |z|^2$ and $dd^c |z|^2$. The last line uses $\operatorname{vol}_{2n}(B_1) = \pi^n / n!$.

Returning to (6), we have, for each regular $s \in (0, 1)$,

$$\int_{U_s} (dd^c u)^n \geq s^n (2\pi k)^n.$$

Sard's theorem gives an increasing sequence of regular values tending to 1. Their sublevel sets exhaust $\{u < 1\}$. Monotone convergence of the nonnegative density implies

$$\int_{\{u < 1\}} (dd^c u)^n \geq (2\pi k)^n.$$

Finally, Cauchy–Binet and the displayed Hessian normalization give

$$(dd^c u)^n = 2^n n! \sum_{|I|=n} |\det Df_I|^2 dV_{2n}.$$

Dividing by $2^n n!$ proves (5). □

7. The boundary probability inequality

Lemma 8. Power maps meet the mass hypotheses

Let $D = \text{int } L_\tau$, $g = F^{-1}$, and

$$\Omega = \{z \in \mathbb{C}^n : \langle a_j, z \rangle \in D \ \forall j\}, \quad h_j(z) = g(\langle a_j, z \rangle), \quad f_k = (h_j^k)_{j=1}^m.$$

For each $k \geq 1$, the map f_k satisfies Lemma 7.

Proof. The rows span, so a basis bounds z whenever its row values lie in the bounded set D . Thus Ω is bounded and open, and contains 0. The functions h_j are holomorphic. They vanish simultaneously only at 0. With $c_0 = g'(0) \neq 0$, their powers have leading terms $(c_0 \langle a_j, z \rangle)^k$, which have no nonzero common zero. If $\sum_j |h_j|^{2k} \leq s < 1$, every row value lies in the compact subset $F(\{|w| \leq s^{1/(2k)}\})$ of D . A basis bounds all such z , and limits retain these inclusions and the sublevel inequality. The sublevel is therefore compact in Ω . $\square$

Theorem 9. Boundary probability bound

For the events E_I of Lemma 5,

$$\sum_I \mu_\tau^{\otimes n}(E_I) \geq 1. \tag{C}$$

Proof. Let $M_{I,k}$ be the integral of $|\det D(f_k)_I|^2$ over $\{|f_k|^2 < 1\}$. Derivative rows are multiples of the corresponding a_j , so dependent selections contribute zero. Lemmas 7 and 8 give $\sum_I M_{I,k} \geq k^n \pi^n / n!$.

For a fixed basis use coordinates $u_i = h_i(z)$, $i \in I$, whose inverse is $z = B_I^{-1}(F(u_i))$. The squared complex Jacobian is the real Jacobian. Dropping the unselected terms from $\sum_j |h_j|^{2k} < 1$ enlarges the transformed domain. Substituting $u_i = \rho_i^{1/k} e^{i\theta_i}$ then gives

$$\frac{M_{I,k}}{k^n} \leq \int_{\rho_i > 0, \sum_i \rho_i^2 < 1, 0 \leq \theta_i < 2\pi} \mathbf{1}\{\langle a_j, B_I^{-1}(F(\rho_i^{1/k} e^{i\theta_i})) \rangle \in D \ \forall j\} \prod_i \rho_i \, d\rho \, d\theta.$$

The factor follows from $k^2 |u_i|^{2k-2} dV_2(u_i) = k \rho_i \, d\rho_i \, d\theta_i$. This fixed measure space has mass $\pi^n / n!$. At every positive ρ_i , the interpolants converge to the boundary interpolant. A limiting row outside $L_\tau = \overline{D}$ eventually violates its constraint. The limsup of the indicator is thus at most the indicator of E_I . Reverse Fatou on this finite space yields

$$\limsup_{k \rightarrow \infty} \frac{M_{I,k}}{k^n} \leq \frac{\pi^n}{n!} \mu_\tau^{\otimes n}(E_I).$$

Summing the finitely many inequalities and combining with the mass lower bound proves (C). $\square$

8. From polytopes to arbitrary convex bodies

Theorem 10. Symmetric Funk volume lower bound

For every symmetric convex body $K \subset \mathbb{R}^n$ and $0 < \tau < 1$,

$$V_K(\tau) \geq \frac{(4 \operatorname{artanh} \tau)^n}{n!}.$$

Proof. For a symmetric polytope P , choose its signed irredundant strip presentation. Combining Theorem 9 and Lemma 6 gives

$$1 \leq \frac{n!}{(4 \operatorname{artanh} \tau)^n} V_P(\tau),$$

which is the required bound.

Now let K be an arbitrary symmetric convex body and fix $0 < \sigma < \tau < 1$. We first construct a symmetric polytope P with

$$K \subset P \subset (\tau/\sigma)K.$$

Put $\lambda = \sigma/\tau < 1$ and choose a finite symmetric set of points in K° whose convex hull Q satisfies $\lambda K^\circ \subset Q \subset K^\circ$. For completeness, such a set is obtained from a sufficiently fine finite net in K° : its support function differs from h_{K° by at most the mesh size on the unit sphere. Since K° contains a ball, choose the mesh size smaller than $(1 - \lambda) \min_{|u|=1} h_{K^\circ}(u)$. Support-function comparison then gives the first inclusion. Take $P = Q^\circ$. This is a bounded symmetric polytope, and polarity gives the required sandwich.

For $x \in \sigma P$ one has $x \in \tau K$ and $K - x \subset P - x$, hence $(P - x)^\circ \subset (K - x)^\circ$. Nonnegativity of the integrand implies

$$\frac{(4 \operatorname{artanh} \sigma)^n}{n!} \leq V_P(\sigma) \leq V_K(\tau).$$

Let $\sigma \uparrow \tau$. Only the scalar expression on the left needs continuity; no limit is interchanged with the body integral. Lemma 2 already gives finiteness of the quantity on the right. $\square$

9. An upper bound in terms of flags

Theorem 11. Flag upper estimate

For every symmetric n -dimensional polytope P ,

$$\limsup_{R \rightarrow \infty} \frac{V_P(1 - e^{-R})}{R^n} \leq \frac{\# \text{Flags}(P)}{(n!)^2}. \quad (7)$$

Proof. For each nonempty proper face choose a point in its relative interior, using the same point whenever the face recurs. For a flag F , write these points as $p_0, \dots, p_{n-1}$ and set $p_n = 0$. The simplices $\text{conv}(p_0, \dots, p_n)$ cover P : triangulate each face recursively by coning its boundary triangulation to its chosen interior point, and cone the boundary of P to 0. Their vertex matrices are nondegenerate by the strict face chain. Use the same construction for P° , with proper-face points $q_0, \dots, q_{n-1}$ along a flag G .

For $x \in \text{int } P$, the projective map

$$T_x(q) = \frac{q}{1 - \langle q, x \rangle}$$

maps P° onto $(P - x)^\circ$. Indeed, for $q \in P^\circ$ the defining polar inequalities prove one inclusion. Conversely, if $y \in (P - x)^\circ$, the bound $h_P(y) \leq 1 + \langle y, x \rangle$ makes the latter denominator positive and puts $y/(1 + \langle y, x \rangle)$ in P° . On a convex combination the projective map reweights its coefficients by positive denominators. It therefore maps each polar flag simplex onto the simplex with vertices $0, T_x(q_0), \dots, T_x(q_{n-1})$.

Using both finite covers and the determinant formula for simplex volume,

$$V_P(\tau) \leq \frac{1}{n!} \sum_{F, G} \int_{D_{F, G}(\tau)} \frac{du_0 \cdots du_{n-1}}{u_0 \cdots u_{n-1}},$$

where $u_i = 1 - \langle q_i, x \rangle$ and $D_{F, G}(\tau)$ is the image of $\tau \text{conv}(p_0, \dots, p_n)$. The Jacobian $|\det(q_i)|$ cancels. Its vertices have coordinates

$$u_i = e^{-R} + (1 - e^{-R})a_{ki}, \quad a_{ki} = 1 - \langle q_i, p_k \rangle, \quad \tau = 1 - e^{-R}.$$

Here $0 \leq a_{ki} \leq 2$, and $a_{ni} = 1$. Set $S_i = \{k : a_{ki} > 0\}$. If a supporting linear functional attains its maximum at a relative-interior point of a face, it is constant on that face. Applying this fact first to the primal flag and then to the polar flag shows that each S_i is a final interval of $\{0, \dots, n\}$ and

$$S_0 \subseteq S_1 \subseteq \cdots \subseteq S_{n-1}.$$

The remainder of the estimate uses only these support relations.

10. The leading coefficient of the flag upper bound

Proof of Theorem 11, continued. Put $z_i = \log u_i$. The integral for a pair (F, G) is $1/n!$ times the ordinary volume of its logarithmic image $Z_{F,G}(R)$. Since $e^{-R} \leq u_i \leq 2$, this image lies in $[-R, \log 2]^n$.

The inclusion $S_i \subseteq S_{i+1}$ supplies a constant $C_i \geq 1$ with $a_{ki} \leq C_i a_{k,i+1}$ for every k . This is a finite coefficient comparison, including zeros. Convex combinations of the vertices then satisfy $u_i \leq C_i u_{i+1}$. Choose positive constants d_i recursively so that $d_{i+1} = C_i d_i$. The shifted coordinates $z_i + \log d_i$ are ordered increasingly and all lie in an interval of length $R + C_{F,G}$, for a fixed sufficiently large $C_{F,G}$. Consequently,

$$\text{vol}_n(Z_{F,G}(R)) \leq \frac{(R + C_{F,G})^n}{n!}. \quad (8)$$

Indeed, the $n!$ coordinate orderings partition a cube up to null hyperplanes and have equal volume.

If $S_i = S_j$ for $i \neq j$, finite coefficient comparison in both directions bounds u_i/u_j above and below by positive constants. Thus $z_i - z_j$ lies in a fixed bounded interval. Slicing the surrounding box gives $\text{vol}_n(Z_{F,G}(R)) = O(R^{n-1})$. If some S_i is all of $\{0, \dots, n\}$, all its coefficients have a positive minimum; hence u_i has a positive lower bound independent of large R . The same degree estimate follows. Constants may depend on this fixed flag pair.

The only remaining case consists of n distinct proper final intervals containing n . They must have cutoffs $n, n-1, \dots, 1$, in this order. Equivalently the pair has the contact pattern

$$\langle q_i, p_k \rangle = 1 \iff k + i < n. \quad (9)$$

For a fixed primal flag, at most one polar flag has this pattern. To see this, fix i and consider the face

$$D_i = \{q \in P^\circ : \langle q, p_{n-i-1} \rangle = 1\}.$$

If (9) holds, relative-interior contact gives $G_i \subset D_i$. Every $q \in D_i$ equals 1 on the whole primal face F_{n-i-1} , and therefore on $p_0, \dots, p_{n-i-1}$. These $n-i$ vectors are linearly independent: affine independence follows successively from the face chain, and their affine span is in a supporting hyperplane avoiding 0. Hence $\dim D_i \leq i$. Since $\dim G_i = i$, the two nested faces are equal. Each G_i is thus forced by F .

There are at most $\#\text{Flags}(P)$ pairs with the contact pattern. Each contributes at most $1/(n!)^2$ to the normalized limsup, by (8) and the preceding factor $1/n!$. All other pairs contribute zero. The number of pairs is finite, so summing these bounds proves (7). $\square$

11. The complete-flag bound

Proof of Theorem 1. Theorem 10 proves the Funk lower bound, and Lemma 2 proves finiteness. Let P be a symmetric n -dimensional polytope and put $N = \# \text{Flags}(P)$. For every $R > 0$, Theorem 10 and (1) give

$$\frac{V_P(1 - e^{-R})}{R^n} \geq \frac{1}{n!} \left(\frac{2 \log(2e^R - 1)}{R} \right)^n.$$

The expression on the right tends to $2^n/n!$. Theorem 11 therefore implies

$$\frac{2^n}{n!} \leq \liminf_{R \rightarrow \infty} \frac{V_P(1 - e^{-R})}{R^n} \leq \limsup_{R \rightarrow \infty} \frac{V_P(1 - e^{-R})}{R^n} \leq \frac{N}{(n!)^2}.$$

Multiplication by $(n!)^2$ yields

$$\boxed{\# \text{Flags}(P) \geq 2^n n!}.$$

Lemma 3 shows that the cube attains the volume bound. It also has exactly $2^n n!$ complete flags: choose one of its 2^n vertices and then an ordering of the n coordinates to be released successively along the flag. Thus the numerical constants in both inequalities are sharp. $\square$